Example 6.11: # program to calculate the sum of numbers 1 to 100

```
n = 100
sum = 0
for counter in range(1,n+1):
        sum = sum + counter
print("Sum of 1 until %d: %d" % (n,sum))
```

Output:

Sum of 1 until 100: 5050

In the above code, *n* is initialized to 100, *sum* is initialized to 0, the *for* loop starts executing from 1, for every iteration the value of sum is added with the value of counter variable and stored in sum. Note that the for loop will iterate from 1 till the upper limit -1 (ie. Value of n is set as 100, so this loop will iterate for values from 1 to 99 only, that is the reason why we have set the upper limit as n+1)

Note

for loop can also take values from string, list, dictionary etc. which will be dealt in the later chapters.

Following is an example to illustrate the use of string in range()

Example 6.12: program to illustrate the use of string in range() of for loop

```
for word in 'Computer':
    print (word,end=' ')
else:
    print ("\nEnd of the loop")
```

Output

```
C o m p u t e r
End of the loop
```

(iii) Nested loop structure

A loop placed within another loop is called as nested loop structure. One can place a **while** within another **while; for** within another **for; for** within **while** and **while** within **for** to construct such nested loops.

Following is an example to illustrate the use of for loop to print the following pattern

```
1
1   2
1   2   3
1   2   3   4
1   2   3   4   5
```